

12345678

UNNI C02 SMARTIFICATION – REV A

Two-sheet cleanup: standard power symbols, grid-aligned wiring, explicit local connectivity.

USB + Power

ESP32–C6 + Auxiliary

Simulation Harness

File: 01\_usb\_power.kicad\_sch

File: 02\_esp\_aux.kicad\_sch

File: 03\_simulation\_harness.kicad\_sch

Unni C02 Smartification

Sheet: /  
File: unni-smartification-c6.kicad\_sch

Title: Unni C02 Smartification – ESP32–C6 Rev A

Size: A3Date: 2026–08–19

KiCad E.D.A. 10.0.5Id: 1/4

A

B

C

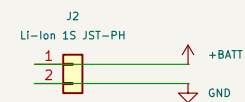
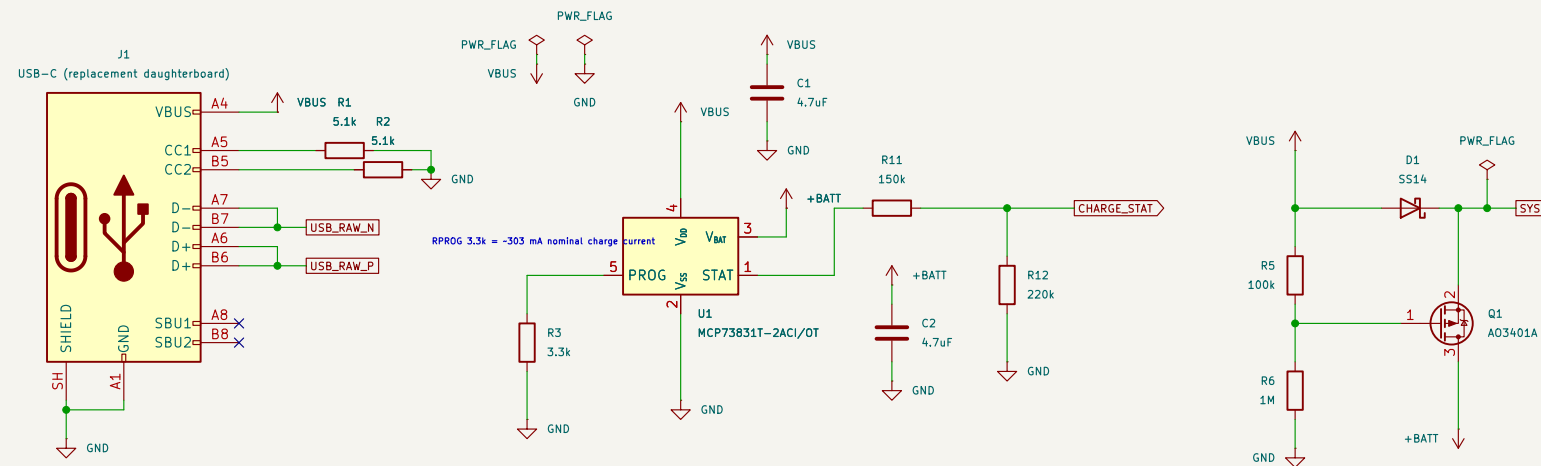
D

E

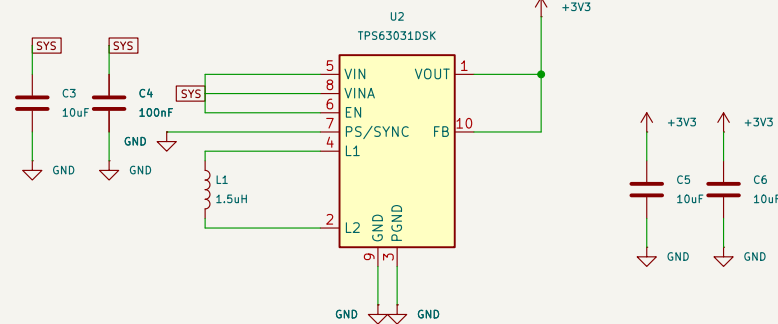
F

## USB + POWER

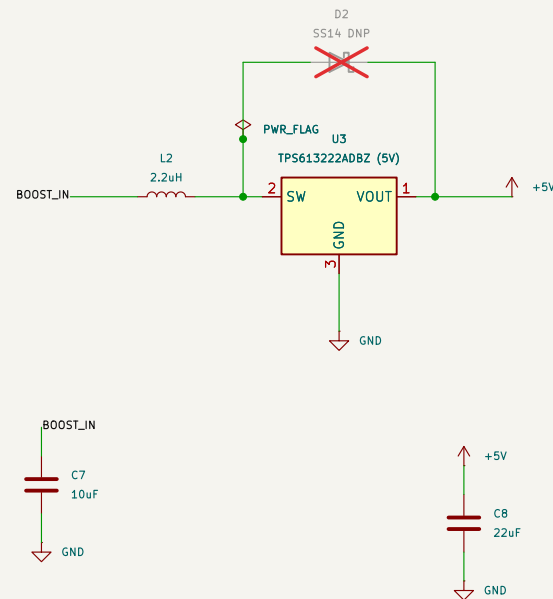
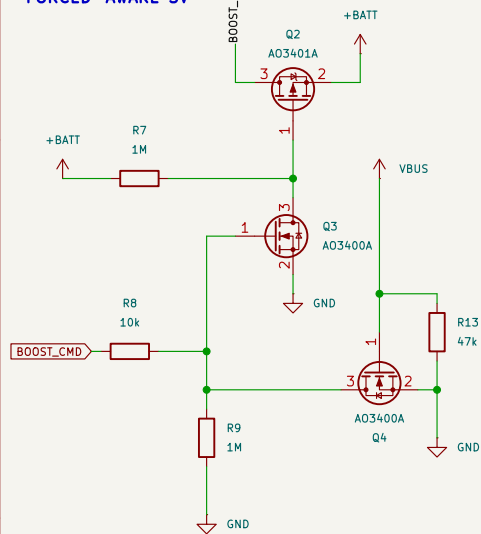
All placement/wiring is on 50 mil (1.27 mm) grid. Standard KiCad symbols used where available.



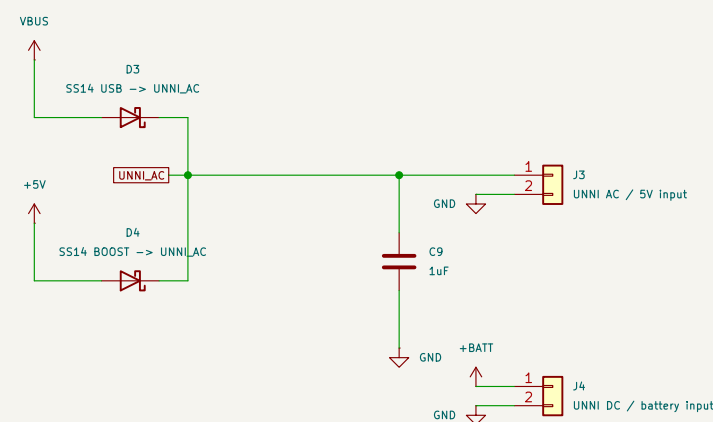
MCP73831 runs autonomously: R3 = 3.3 kΩ sets -300 mA; STAT is monitored by the ESP. Battery NTC remains firmware-supervised.



FORCED-AWAKE 5V



## UNNI POWER OUTPUTS



## Unni CO2 Smartification

Sheet: /USB + Power/  
File: 01\_usb\_power.kicad\_sch

**Title: USB + Power**

|          |                  |
|----------|------------------|
| Size: A3 | Date: 2026-08-19 |
|----------|------------------|

|                     |  |
|---------------------|--|
| Size: A5            |  |
| KiCad E.D.A. 10.0.5 |  |

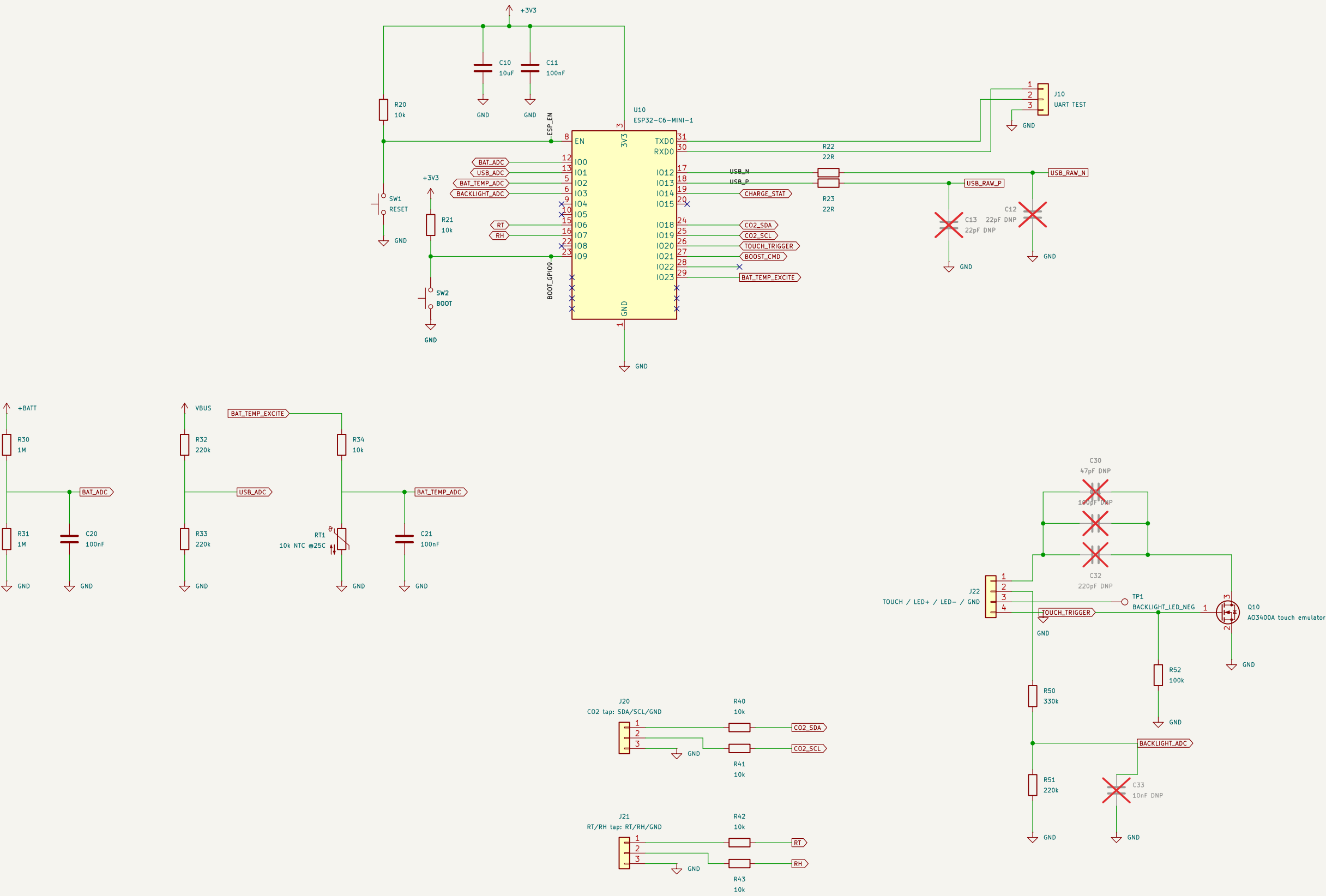
Date: 2026-08-19

Rev: A-v29-validated

Id: 2/4

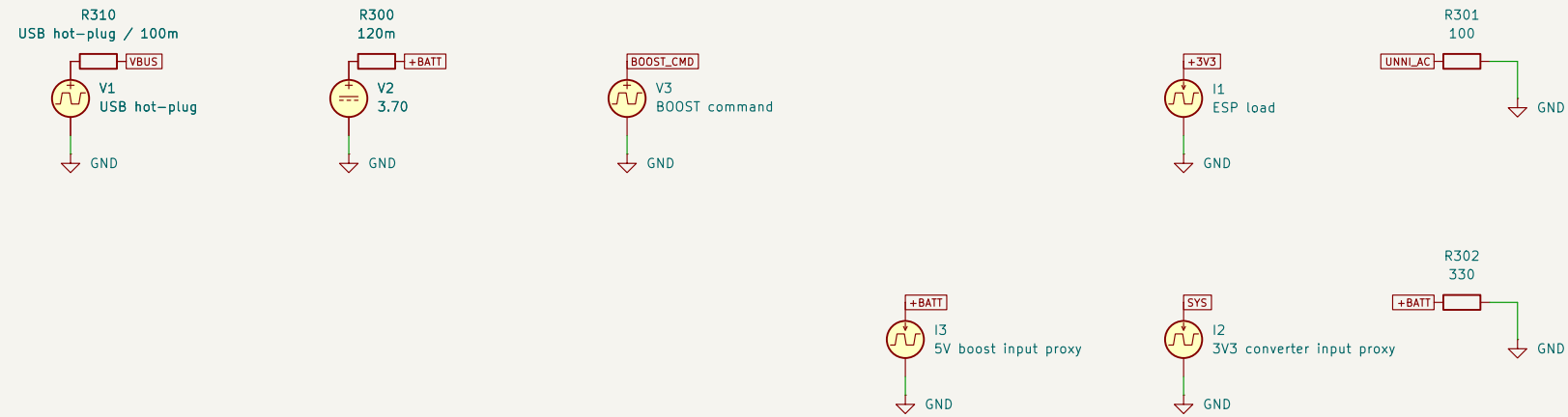
ESP32-C6 + AUXILIARY CIRCUITS

GPIO0-3 are reserved for ADC inputs; RT/RH use RTC-capable GPIO6/7 for deep-sleep wake.



**SIMULATION HARNESS – excluded from BOM/PCB**

Scenario: battery-only → forced awake → USB hot-plug while BOOST\_CMD remains asserted → autonomous charging → USB removal. Battery model includes 120 mΩ ESR. Battery model includes 120 mΩ ESR. I2 is the 3V3-converter input-power proxy. TPS613222A input power/discharge is modeled internally; legacy I3 is excluded from simulation.



## Unni CO2 Smartification

Sheet: /Simulation Harness/  
File: 03\_simulation\_harness.kicad\_sch

**Title: Simulation Harness**

|          |                  |
|----------|------------------|
| Size: A3 | Date: 2026-08-19 |
|----------|------------------|

Rev: A-v29-validated

Id: 4/4